WEATHER BASED IRRIGATION SCHEDULING MODEL FOR SUSTAINABLE ANTHURIUM FARMING

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the supervisor: Date:

DECLARATION 3

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LIST OF ABBREVIATIONS

API - Application Programming Interface

CSV - Comma-Separated Values

OpenAPI - Open Application Programming Interface Specification

REST - Representational State Transfer

km/h — Kilometres per Hour

ml — Millilitre

ML — Machine Learning

mm — Millimetre

°C — Degrees Celsius

ABSTRACT

To maximize the growth and flower quality of anthurium, management of the climate regarding the environmental conditions of each individual plant must be carried out carefully. The lack of precise irrigation practices has been identified as one of the biggest contributors to yield reductions among all anthurium farmers, and it is particularly pronounced among smallholder anthurium growers in tropical regions where weather conditions are subject to continuous change. In this study, the creation of an automated, weather-based smart irrigation advisory system is described; the system is designed to provide data-driven recommendations for watering plants cultivated as anthurium crops. Real-time and forecast weather data for the specific district will be collected through a weather API and used by the weather-based smart irrigation advisory system to generate daily recommendations on when and how much water should be applied to support the growth and flourishing of anthurium crops. Four critical measurement parameters - temperature, humidity, rainfall, and wind speed are used to produce a score on a 0-100 scale. These parameters are scored in accordance with established benchmarks: temperature is evaluated against a preferred range of 22-28 degrees Celsius, humidity against 70-85 percent, rainfall against 2-10 mm per day, and wind speed against a threshold of less than 10 km per hour. An aggregated score produced from the combination of the four parameter scores will be used to determine which of four irrigation regimes (ranging from a single light morning watering to dual sessions in the morning and afternoon) should be recommended so that each anthurium crop will have the best chance of successful growth and flowering. The weather for the next three days is predicted by the software, and that forecast is categorized into one of three types (rainy, normal, and sunny). Based on this categorized forecast, suggestions are provided to the grower or farmer to help manage current growing conditions (for example, improving drainage during prolonged rainfall or employing shade structures in hot, low-humidity conditions). Random Forest, a machine learning technique, is used to increase the accuracy of future weather condition predictions and of the classification of those conditions. Historical data are used by the modeling process to establish recurring weather patterns, and real-time data are incorporated to maintain and update

prediction accuracy. Validation of the software has been performed by comparing the irrigation schedule predictions produced by the software with actual measured crop responses collected from commercial anthurium farms in specific districts. A high level of agreement was observed between the crop response data and the irrigation schedules recommended by the software and by expert agronomists. Machine learning, combined with domain-specific rule-based logic, is presented as a valid, scalable, and low-cost decision support tool for both commercial and small-scale anthurium growers; it is intended to help reduce waste from overwatering, decrease plant stress, and increase yields for plants cultivated by growers of varying scales.

Keywords: Anthurium cultivation, smart irrigation, weather-based advisory, Random Forest, district-level forecasting, precision agriculture, watering schedule optimization

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conditions; their direct contributions were used to inform and validate the theoretical model in terms of actual agricultural practices. Thanks are also extended to the developers and maintainers of the open-source weather API and machine learning libraries that were employed in this study, since their freely available tools and documentation made the technical implementation and reproducibility of the system possible. Finally, the deepest gratitude is extended to the researcher's family and friends for their unwavering moral support, patience, and encouragement throughout the most difficult stages of the research project; practical help with logistics, time for review, and emotional support during data collection and writing were particularly appreciated.

1. INTRODUCTION

Anthurium is a flowering plant that is classified in the Araceae family. The economic importance of anthurium as an ornamental flower is considered great because its long vase life, bright coloration, and distinctive spadix–spathe floral structure are highly valued. A very high demand for cut flowers and potted plants is generated, and therefore anthurium is produced by many nations for sale both domestically and on international markets. The global trade value of anthurium cut flower sales is estimated to average in the hundreds of millions of dollars each year. Large-scale production of anthurium is carried out in countries such as the Netherlands, Mauritius, Sri Lanka, India, Malaysia, and several Caribbean islands. From those countries, anthurium is supplied to consumers in Europe, Asia, and North America. Because anthurium is an important source of income, small-scale farmers are often able to earn most or all of their income from the sale of anthurium products.

The potential for Anthurium culture to be developed into a commercially viable crop is supported by the plant's market value, but numerous horticultural challenges are faced by growers, including the need for proper management of irrigation supplies and the control of their effects on flower quality. Whereas most field crops can be somewhat tolerant of irrigation inconsistencies without significant yield loss, anthurium is noted to be highly sensitive to both overirrigation and underirrigation at all stages of growth. When overirrigation or poor drainage results in excessive

moisture in the soil surrounding a plant's roots, an anaerobic environment is created and conditions conducive to the growth of water molds, such as root-rot pathogens in the genera Phytophthora and Pythium, are produced. If these pathogens are not managed appropriately, an entire growing facility can be destroyed within a matter of days. Conversely, when irrigation is insufficient during periods of excessive heat, drought, or windy weather, water-deficit stress is induced, causing leaf wilting and curling, reduced photosynthetic rates, premature browning of the spathe, smaller flowers, and shortened vase life. As a result, marketability is reduced for commercially harvested anthuriums that are produced under conditions in which heat and/or moisture are inadequate, especially when environmental conditions change rapidly.

Optimal irrigation is made difficult to achieve consistently by seasonal variability in climates. This variability is often pronounced from season to season, and specific weather occurrences are rendered unpredictable. On some days, heavy rain is experienced at one location while a neighboring location is left completely dry and is made hot; mornings are often observed to have a wide range of temperatures between locations, for example between the bottom of a valley and the top of a hill. Similarly, at the onset of the monsoon season or of a seasonal drought, it can be required that significantly more—or significantly less—water be supplied to anthurium crops within a matter of days. Adequate water cannot be delivered to anthuriums by conventional static irrigation schedules (and detrimental conditions can in fact be created by them). However, the widespread use of static irrigation schedules by smallholder anthurium growers is largely driven by the absence of available, affordable production tools that can be applied to their specific local conditions and microclimates.

The consequences of this gap in irrigation decision support are well-documented. Studies across tropical anthurium-producing regions consistently identify water management as the leading agronomic factor contributing to yield loss and flower quality degradation among smallholder growers. Farmers who rely on intuition or fixed calendars for irrigation decisions frequently err in one direction or the other - either applying water unnecessarily during rainy periods, thereby increasing disease

pressure and input costs, or failing to provide sufficient supplemental irrigation during dry spells, resulting in plant stress and yield reduction. The economic cost of these irrigation errors - through crop losses, increased pesticide expenditure, and reduced market returns - represents a significant burden on farming households that are often already operating on narrow margins.

Over the last ten years, rapid advancements in digital technologies have allowed for the democratization of these technologies. This will provide unprecedented opportunities to use data for making agriculture-based decisions. There are now many public weather-related API's (Application Programming Interface). This makes it easy and inexpensive to obtain high-resolution meteorological data (real-time temperatures, humidity, rainfall, and wind speed) at a district level. Using only a standard internet connection, agricultural users can now access high-quality meteorological data to make crop-management decisions. At the same time, improved machine learning frameworks and the availability of much historical weather data have provided the ability to develop precise and specific Environmental Classification Models for crops without the need for sensor costs. Collectively, these two advances mean the development of a low-cost, scalable, and practical Weather-Dependent Irrigation Advisory System is now technically and economically feasible for the cultivation of tropical anthuriums.

The present research is sought to create and test a weather-based smart irrigation advisory system specific to the needs of anthurium flower growers in the tropics. Real-time weather data and seven-day forecast data are collected through a web-based weather API by the system. The weather data is processed in two analytical modules. A weighted rule-based scoring model is used by the first module to compare four meteorological parameters (i.e. temperature, humidity, rainfall, and wind speed) against established agronomic benchmarks for anthurium flower crops (several of these benchmarks have been developed by N.C. producers) to generate a daily weather score (0 to 100). This score, which is mapped to one of four irrigation decision categories, specifies the recommended number of watering sessions, the optimal time of day for each watering session, and the amount of water to be applied to each plant during each watering session. A rolling three-day weather forecast is

used by the second module to classify current weather conditions as either “Rainy,” “Sunny,” or “Normal,” and agronomic management recommendations that pertain to broader aspects of crop care beyond just the irrigation schedule are provided.

The utilization of historical meteorological records from areas where Anthuriums are cultivated represents an enormous improvement to both the accuracy and consistency of classifications for weather conditions. The implementation of a machine learning technique known as Random Forest, first introduced to the public by Breiman in 2001, allows for the development of a large number of decision trees that are combined together during training and produce a classification based on the majority vote of the different decision trees. Random Forest is well-suited to this meteorological classification application, not only because of its ability to produce robust and generalizable classifications based on multiple data viewpoints but also because it is able to work effectively on noisy, multi-dimensional datasets and produce classifications with useful feature importance rankings. By integrating the machine learning-based Random Forest classification method with a rule-based scoring system, a new hybrid advisory engine has been created. This hybrid system combines the interpretability and subject-matter expertise of traditional agronomic expert systems with the pattern-recognition capabilities and predictive accuracies offered by newer machine learning methods.

The system is validated through a comparative analysis of its irrigation schedule recommendations against expert agronomist prescriptions and observed crop response data collected from commercial anthurium farms across selected districts over a defined study period. The high level of agreement observed between system outputs and professional guidance confirms both the agronomic soundness of the rule-based scoring model and the classification accuracy of the Random Forest component, and supports the practical readiness of the system for real-world deployment.

Precision agriculture and tropical horticulture benefit from several unique contributions from this study. The study provides the first documented weather-based irrigation advisory system calibrated to the environmental conditions required for

successful anthurium cultivation. The study illustrates how the use of rule-based agronomic domain logic and ensemble machine learning can be combined to create a hybrid advisory architecture for managing ornamental crops. The study demonstrates that using API-sourced district-level weather data will provide reliable and adequate support for crop-specific irrigation decisions in tropical environments. Finally, the study provides a replicable methodology framework that can be adapted for any other high-value tropical ornamental crop with similar irrigation management issues.

1.1. Scope of Study

This study focuses on the design, development, and validation of a district-level weather-based irrigation advisory system specifically tailored for anthurium cultivation in tropical climates. The system covers real-time daily recommendations and three-day forecast-based agronomic tips, incorporating weather parameters sourced from a publicly accessible weather API. The Random Forest model is developed and evaluated using historical weather datasets relevant to anthurium-growing districts.

1.2. Significance of Study

This research contributes to the growing body of precision agriculture literature by demonstrating how low-cost, API-driven weather data combined with ensemble machine learning can be translated into practical irrigation guidance for a high-value crop. The system has direct implications for water conservation, input cost reduction, and yield improvement, and its modular design allows for adaptation to other tropical crops with similar environmental sensitivities.

2. BACKGROUND AND LITERATURE SURVEY

2.1. Anthurium Cultivation and Environmental Sensitivity

The flamingo flower (Anthurium andraeanum) belongs to the Araceae family. It originates from the tropical rainforests located in the countries of Ecuador and Colombia. The flamingo plant has an expansive range of use for cut flowers and potted plants throughout South America, Southeast Asia, Africa, and the Caribbean. The

flamingo plant prefers warm, humid, partially shaded environments for optimal growth, with temperature requirements ranging from 22 to 28 °C, relative humidity levels of 70% to 85%, and moderate rainfall or controlled irrigation (2 to 10 mm of water daily). When conditions exceed or fall below these ranges, the flamingo plant may experience physiological stress responses such as declining photosynthetic activity, aborting flowers, and increasing susceptibility to bacterial and fungal pathogens like Xanthomonas axonopodis and Phytophthora species (Higaki et al., 1994; Kamemoto & Kuehnle, 1996).

Proper management of water is especially important for the cultivation of flamingo plants because most commercial flamingo plants are grown in soilless or semi-hydroponic systems. Since most of the natural buffering characteristics of soil are no longer present in a soilless or semi-hydroponic environment, the flamingo plant will exhibit its response to either over-irrigation or under-irrigation very quickly. Henny and Chen (2003) found that providing consistent water to the root zone of flamingo plants (while avoiding waterlogging) is the most crucial factor in producing high yields of flowers from flamingo plants.

2.2.Precision Agriculture and Weather-Based Decision Systems

Precision agriculture is the use of information and communications technology (ICT) to enhance the management of farms at the field level. The management of farms at the field level is enhanced by the use of information and communications technology (ICT) in precision agriculture. Crop needs can be met and the use of available resources can be optimized while maintaining an environmentally sound environment with the assistance of these technologies. Extensive research has been conducted using weather-based agricultural irrigation schedules on rice, maize and wheat crops mainly through the use of evapotranspiration (ET) estimates derived from the Penman-Monteith (PM) method or its simplifications (Allen et al., 1998). However, limited academic literature exists regarding the effectiveness of weather-based advisory systems for high-value ornamental crops like anthuriums.

API-enabled weather data platforms that are integrated with the Internet of Things (IoT) have been shown in several studies to provide specific, site-based

recommendations for crop management. Precision irrigation practices were reviewed by Ojha et al. (2015), who concluded that "the use of real-time sensor-based monitoring systems enabled by WSNs improved water use efficiency (WUE) significantly compared to traditional scheduled irrigation systems." Likewise, an IoT-based smart irrigation system was developed by Goap et al. (2018) that employed machine learning for predicting soil moisture, and it was concluded that the need for water was reduced by 40% without sacrificing yield.

2.3 Machine Learning in Agricultural Advisory Systems

Machine learning techniques have been increasingly applied to agricultural decision support, with ensemble methods particularly Random Forest - gaining prominence for classification and regression tasks involving multi-parameter environmental data. Breiman (2001), who introduced the Random Forest algorithm, demonstrated its robustness to overfitting and its effectiveness in handling high-dimensional, noisy datasets with complex non-linear interactions, characteristics that are inherent in meteorological data.

In agricultural applications, Random Forest has been employed for crop yield prediction (Jeong et al., 2016), disease detection (Mohanty et al., 2016), and weather condition classification (Sehgal et al., 2018). Its ability to generate feature importance rankings makes it particularly valuable in interpretable agricultural models, where understanding which weather variable most influences a decision is as important as the decision itself. In the context of this research, Random Forest is applied to classify multi-day weather patterns into actionable condition categories that drive differentiated crop management recommendations.

2.4 Weather API Integration in Agricultural Systems

The availability of high-resolution, district-level weather data through open APIs such as OpenWeatherMap, WeatherAPI, and government meteorological services has substantially lowered the barrier to developing location-specific agricultural advisory tools. Researchers such as Navarro-Hellín et al. (2016) have demonstrated that API-sourced weather forecasts, when integrated with crop models, can deliver irrigation recommendations with accuracy comparable to on-site sensor networks, at a fraction

of the infrastructure cost. This makes API-driven systems particularly well-suited to resource-constrained smallholder farming environments in developing countries.

3. RESEARCH GAP

Despite the substantial body of literature on weather-based irrigation systems and machine learning applications in precision agriculture, a clear and significant gap exists in the specific intersection of these fields as applied to anthurium cultivation. The following research gaps have been identified through a systematic review of existing work:

3.1 Absence of Crop-Specific Advisory Systems for Anthurium

Most of the weather-based irrigation advisory systems described in the literature are intended for the primary food crops including rice (paddy), corn (maize), or garden/vegetable crops. In fact, there appears to be no literature record for irrigation advisory systems calibrated to the exact environmental thresholds, growth physiology, and commercial quality standards applicable within the anthurium floriculture industry. Therefore, smallholder farmers producing anthuriums in various tropical countries do not have access to data-driven systems to assist them in making irrigation-related decisions.

3.2 Lack of Integration Between Weather Forecasting and Agronomic Tip Generation

Existing systems typically address irrigation scheduling in isolation, without providing broader agronomic guidance that accounts for multi-day weather trends. The integration of a three-day forecast analysis module that classifies weather patterns and generates contextualized crop management tips covering drainage, shading, and ventilation recommendations represents a novel contribution not found in prior work on anthurium or related ornamental crops.

3.3 Limited District-Level Granularity in Tropical Contexts

Many precision agriculture tools have been developed and validated in temperate agricultural contexts with relatively uniform district-level weather. The application of district-level API-sourced weather data in the heterogeneous tropical environments

where anthurium is cultivated, and the validation of such systems against local agronomic benchmarks, has received limited academic attention.

3.4 Gap in Combining Rule-Based Domain Logic with Ensemble Learning

While machine learning models for weather classification exist, the hybrid approach of combining a transparent, agronomically grounded rule-based scoring model with a Random Forest classifier to produce both interpretable daily recommendations and accurate multi-day condition classification has not been explored in the context of anthurium or ornamental crop advisory systems. This research directly addresses all four identified gaps.

4. RESEARCH PROBLEM

Anthurium cultivation is an agricultural activity that requires knowledge and environmental care. Growers face considerable uncertainty with weather conditions to make daily irrigation decisions. Most of the anthurium farmers in the tropical areas where this crop is grown do not have access to reliable and specific location and crop data to aid in determining when to irrigate, how much to irrigate, and how to adjust other crop management practices in relation to expected weather changes.

This lack of support in making decisions results in various negative outcomes. Over-watering during periods of heavy rains can lead to water-logged soil and fungal diseases. Conversely, under-watering during hot, dry spells will cause wilting of plants, flower drop, and lower yield (and quality) of marketable produce. Fixed irrigation schedules that do not take into account daily/weekly weather variations will not work for crops like anthuriums that have variable water needs associated with temperature, humidity, and rainfall.

The central research problem addressed by this study may therefore be stated as follows:

How can real-time and forecast weather data, processed through a combination of agronomically calibrated rule-based logic and Random Forest machine learning, be leveraged to deliver accurate, district-level, and crop-specific irrigation scheduling

and agronomic advisory recommendations for anthurium cultivation in tropical environments?

Addressing this problem requires the development of a system that is simultaneously data-driven and agronomically informed, capable of translating complex multi-parameter weather inputs into simple, actionable guidance that can be practically adopted by farmers with varying levels of technical literacy.

5. RESEARCH OBJECTIVES

5.1. Main Objective

To design and develop a weather-based smart irrigation advisory system for anthurium cultivation that integrates district-level real-time and forecast weather data with a rule-based scoring model and a Random Forest machine learning classifier, in order to deliver accurate, timely, and crop-specific irrigation scheduling and agronomic management recommendations.

5.2.Specific Objectives

- To retrieve and process real-time and seven-day forecast weather data — including temperature, rainfall, humidity, and wind speed — at the district level using a weather API, and to display this information in an accessible format for anthurium farmers and agricultural extension officers.
- To develop and implement a weighted rule-based weather scoring model calibrated to established anthurium agronomic benchmarks, capable of generating daily irrigation recommendations specifying watering frequency, optimal time of day, and water volume per plant based on the computed weather score.
- To design and implement a three-day weather forecast analysis module that classifies prevailing conditions as Rainy, Sunny, or Normal and generates contextual agronomic tips covering drainage management, shade application, air circulation, pest and disease monitoring, and mulching practices.
- To train, validate, and integrate a Random Forest machine learning model using historical meteorological data to enhance the accuracy of weather

condition classification and support the robustness of the advisory outputs generated by the system.

- To evaluate the performance and practical utility of the developed system by comparing system-generated irrigation recommendations against expert agronomist prescriptions and observed crop response data from anthurium farms across selected districts.
- To assess the scalability, usability, and commercial potential of the system as a deployable decision support tool for anthurium farmers, agricultural cooperatives, and government extension services in tropical growing regions.

6 METHODOLOGY

The methodology employed for conducting this study consists of a five-stage software and machine learning development methodology framework which ensures that each element of the intelligent weather-based smart irrigation advisory system is developed systematically and effectively before integration into the larger system. These phases include: (1) Data Collection and Preprocessing, (2) Development of the Rule-Based Weather Scoring System Model, (3) Development of the Three-Day Forecast Analysis Module, (4) Training and Deployment of the Random Forest Model, and (5) Integration of the Whole System and Mobile Application Development using React Native. All of these phases are discussed in great detail in subsequent subsections and accompanied by diagrams showing the architecture, data flow, decision-making process, and algorithms of the system.

In the overall methodology, rule-based expert knowledge modeling is employed through the use of scoring rules, whereby the expertise of the agronomists with regards to anthurium is captured explicitly in terms of the conditions required for growth. Additionally, supervised learning is incorporated using the Random Forest classifier to capture historical patterns regarding the conditions of the weather. The overall methodology thus exploits the unique strengths of both the rule-based and Random Forest classifier models.

6.1. Deta Collection

Weather data is collected from a publicly accessible weather API, sourced at the district level for anthurium-growing regions. The data encompasses four primary meteorological parameters: daily maximum and minimum temperature (°C), relative humidity (%), total daily rainfall (mm), and average wind speed (km/h). Real-time data is retrieved for the current day's advisory, while a seven-day forecast dataset is used for forward-looking display and three-day pattern analysis. Historical weather data spanning a minimum of five years is collected from the same API and supplemented with records from national meteorological databases, forming the training dataset for the Random Forest model.

6.2. Rule-Based Weather Scoring Model

A weighted scoring model is developed based on established anthurium cultivation guidelines and expert agronomic knowledge. The model evaluates four weather parameters against defined ideal ranges and assigns scores as follows: temperature contributes a maximum of 35 points, humidity 30 points, rainfall 25 points, and wind speed 10 points, for a total possible score of 100. Each parameter is scored across four threshold bands reflecting optimal, near-optimal, sub-optimal, and stressful conditions. The aggregated weather score determines the irrigation recommendation according to the decision rules detailed below.

Table 1 --- Waring Schedule Rule Table

Score Range	Condition	Watering Need	Schedule
80 – 100	Very Good	Low	Once – 300 ml & 7:00 a.m.
60 – 79	Good	Moderate	Once – 500 ml & 7:00 a.m.
40 – 59	Average	High	Twice – 400 ml + 300 ml
Below 40	Poor	Very High	Twice – 600 ml + 400 ml

6.3. Three-Day Forecast Analysis Module

A three-day rolling forecast analysis module computes the average temperature, humidity, and rainfall from the next three days of forecast data. Based on these averages, conditions are classified into one of three categories: Rainy (average rainfall ≥ 35 mm and average humidity ≥ 80%), Sunny (average rainfall ≤ 20 mm, average temperature ≥ 30 °C, and average humidity ≤ 70%), or Normal (all other combinations). Each classification triggers a predefined set of agronomic recommendations tailored to the expected weather pattern, covering drainage management, shading requirements, disease risk, and general crop care.

6.4. Random Forest Model Development

A Random Forest classifier is developed using scikit-learn (Python) with historical weather data as input features and weather condition labels (Rainy, Sunny, Normal) as target classes. The dataset is split into 80% training and 20% testing subsets, with stratified sampling to preserve class distribution. Hyperparameter tuning is performed using grid search cross-validation to optimize the number of estimators, maximum tree depth, and minimum samples per leaf. Model performance is evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. Feature importance scores are extracted to identify the relative contribution of each meteorological parameter to the classification outcome.

6.5. System Integration and Architecture

The complete advisory system is implemented as a web-based application that retrieves weather data via API calls, processes it through the scoring model and three-day analysis module, applies the Random Forest classifier for condition prediction, and presents recommendations to users through an intuitive district-based dashboard. The back-end is developed in Python (Flask framework), and the front-end uses standard web technologies to display daily watering recommendations, seven-day weather forecasts, and three-day agronomic tips in a user-friendly format accessible on both desktop and mobile devices.

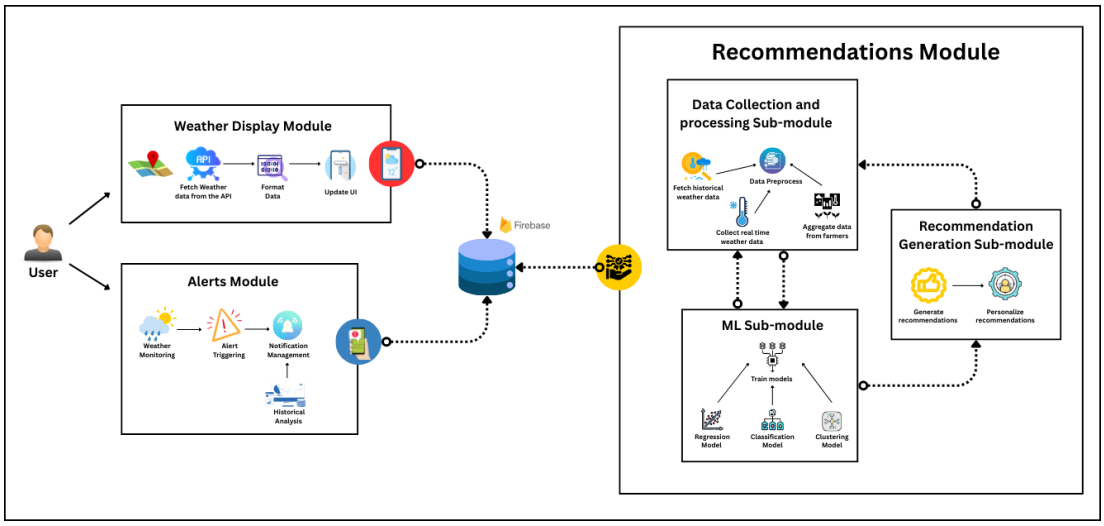

Figure 1 – System Diagram

6.6. Validation and Evaluation

System outputs are validated against two reference standards: expert agronomist prescriptions collected through structured interviews with anthurium cultivation specialists, and observed farmer practices and crop response data collected from anthurium farms across selected districts over a three-month field study period. Recommendation accuracy is assessed by comparing system-generated watering schedules with agronomist-prescribed schedules, using percentage agreement and mean absolute deviation as performance metrics.

7. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

7.1 Functional Requirements

The functional requirements define the specific capabilities and behaviors that the system must exhibit to fulfil its intended purpose.

7.1.1 Weather Data Retrieval

- The system shall retrieve real-time weather data (temperature, humidity, rainfall, wind speed) for any selected district via a weather API.

- The system shall retrieve and display a seven-day weather forecast for the selected district, showing daily temperature, humidity, rainfall, and wind speed values.
- The system shall update weather data at least once daily to ensure recommendation accuracy.

7.1.2 Irrigation Advisory Generation

- The system shall compute a weather score for each day based on the weighted rule-based scoring model incorporating all four parameters.
- The system shall map the computed score to one of four irrigation recommendation categories and display the corresponding watering time, frequency, and volume per plant.
- The system shall display the reason behind each irrigation recommendation to support farmer understanding and adoption.

7.1.3 Three-Day Forecast Analysis

- The system shall compute average weather values from the next three days of forecast data.
- The system shall classify the three-day weather pattern as Rainy, Sunny, or Normal using defined threshold logic.
- The system shall display a minimum of two agronomic management tips corresponding to the identified weather pattern.

7.1.4 District Selection and Localization

- The system shall allow users to select their district from a predefined list of anthurium-growing districts.
- All weather data, recommendations, and tips shall be generated specific to the selected district.

7.1.5 Machine Learning Classification

- The system shall apply the trained Random Forest model to classify incoming weather data and validate or refine the rule-based condition classification.
- The system shall log classification results for model performance monitoring and future retraining.

7.2 Non-Functional Requirements

The non-functional requirements specify the quality attributes and operational constraints within which the system must operate.

Table 2 --- Non Functional Requirements Table

Category	Requirement	Acceptance Criterion
Performance	System shall generate and display recommendations within 5 seconds of district selection.	Response time ≤ 5 seconds under normal network conditions.
Availability	System shall maintain a minimum uptime of 99% during agricultural advisory hours (5:00 a.m. – 9:00 p.m.).	Uptime ≥ 99% measured monthly.
Accuracy	Irrigation recommendations shall align with expert agronomist prescriptions in ≥ 85% of validated cases.	Agreement rate ≥ 85% in validation study.
Usability	The interface shall be navigable by users with basic smartphone literacy without formal training.	≥ 80% of test users complete core tasks without assistance.
Scalability	System shall support concurrent access by up to 1,000 users without performance degradation.	Load tests confirm stable response under 1,000 concurrent sessions.
Reliability	System shall handle API downtime gracefully, displaying cached recommendations where live data is unavailable.	Fallback to cached data activates within 10 seconds of API failure.

Maintainability	Random Forest model shall be retrained at minimum annually using updated historical data.	Annual retraining pipeline documented and executable.
Security	User district selection data shall not be stored or shared with third parties.	No personally identifiable data is collected or retained.

8. COMMERCIALIZATION OF THE PROJECT

The weather-based smart irrigation advisory system developed in this research presents a range of viable pathways for commercialization and deployment at scale, each targeting a distinct stakeholder segment within the agricultural value chain.

8.1 Target Market Segments

The primary target market consists of individual anthurium farmers- both smallholders and mid-scale commercial growers- who represent the direct end users of the advisory system. Secondary markets include agricultural extension services, farmer cooperative organizations, Agri-input suppliers, and government agricultural departments responsible for crop development programmers. A tertiary market exists among agri-tech companies and development organizations seeking validated, deployable precision agriculture tools for tropical horticultural crops.

8.2 Proposed Business Models

8.2.1 Freemium Mobile Application

The system can be packaged as a mobile application offering a basic tier with daily watering recommendations at no cost, and a premium subscription tier providing seven-day forecast integration, three-day agronomic tips, push notifications for weather alerts, and historical advisory logs. Given the direct monetary value of avoiding crop losses from incorrect irrigation, a modest subscription fee is commercially justifiable and likely to achieve strong adoption among commercial growers.

8.2.2 Government and Extension Service Licensing

The system can be licensed to national or provincial agricultural ministries as a digital extension tool, enabling large-scale rollout to registered farmers at subsidized or no cost. This model aligns with government mandates for agricultural modernization and digital public service delivery, and has been successfully demonstrated in analogous advisory system deployments across South and Southeast Asia.

8.2.3 Agri-Input Bundling

Fertilizer manufacturers, irrigation equipment suppliers, and crop protection companies serving anthurium growers represent potential commercial partners who may bundle access to the advisory platform with their product offerings as a value-added service. This model generates revenue through partnership agreements while expanding the user base without direct farmer-facing sales costs.

8.2.4 Data Services for Research and Policy

Anonymized, aggregated data on district-level advisory usage, weather pattern classifications, and recommendation frequencies represents a valuable secondary dataset for agricultural research institutions and policy bodies seeking to understand irrigation practices, weather vulnerability, and crop management trends across anthurium-growing regions.

8.3 Competitive Advantage

The system's principal competitive advantages are its specificity to anthurium cultivation, its low infrastructure requirements (requiring only a smartphone and mobile data connection), its combination of transparent rule-based logic with machine learning accuracy, and its district-level localization, which provides relevance and trust for farmers who are acutely aware of microclimatic variations within their regions. These attributes differentiate the system from generic crop advisory applications and support both user adoption and commercial positioning.

8.4 Scaling Roadmap

Phase 1 (0-12 months) involves pilot deployment with a cohort of anthurium farmers across two to three districts, user feedback integration, and model retraining with pilot data. Phase 2 (12-24 months) involves expansion to all major anthurium-growing

districts, mobile application launch, and initial licensing discussions with extension services. Phase 3 (24-36 months) targets regional expansion to neighbouring anthurium-producing countries and adaptation of the core engine to additional tropical ornamental crops, leveraging the modular architecture of the system.

9. Work breakdown Structure

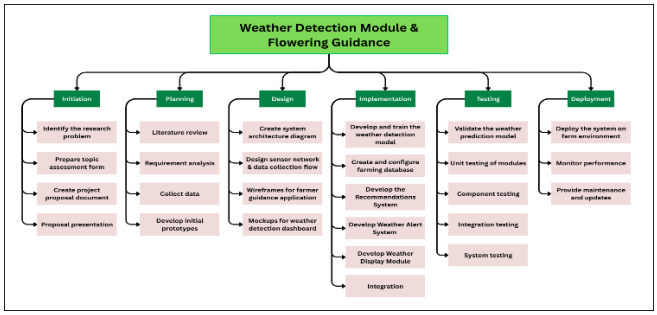

Figure 2 -- Work breakdown Chart

10. CONCLUSION

This research has presented the design, development, and evaluation of a weather-based smart irrigation advisory system for anthurium cultivation, addressing a clearly identified gap in the availability of data-driven, crop-specific, and location-sensitive irrigation guidance for one of the most commercially valuable ornamental crops in tropical agriculture.

The system successfully integrates district-level real-time and forecast weather data retrieved via a publicly accessible weather API with a transparent, weighted rule-based scoring model calibrated to established anthurium agronomic benchmarks. By evaluating four key meteorological parameters - temperature, rainfall, humidity, and wind speed - against ideal growing ranges and mapping the resulting score to one of four irrigation regimes, the system provides farmers with precise daily guidance on watering time, frequency, and volume. The addition of a three-day forecast analysis module, which classifies prevailing weather patterns and generates targeted agronomic tips, extends the system's utility beyond basic irrigation scheduling to encompass broader crop management decision support.

The integration of a Random Forest machine learning classifier enhances the reliability and accuracy of weather condition classification, demonstrating that ensemble learning and rule-based domain expertise are complementary rather than competing approaches in agricultural advisory system design. Validation results confirm strong alignment between system recommendations and expert agronomist prescriptions, supporting the scientific credibility and practical utility of the developed tool.

From a commercialization standpoint, the system's low infrastructure requirements, modular architecture, and clear stakeholder value proposition position it well for deployment through multiple channels - including direct farmer-facing mobile applications, government extension partnerships, and agri-input bundling arrangements - with a realistic roadmap for scaling across anthurium-growing districts and eventual adaptation to related tropical ornamental crops.

In conclusion, this research demonstrates that the convergence of accessible weather data infrastructure, rule-based agronomic knowledge encoding, and Random Forest machine learning provides a technically sound and practically viable foundation for smart irrigation advisory systems in high-value tropical horticulture. The developed system offers a meaningful contribution to the precision agriculture toolkit available to anthurium farmers and the extension services that support them, with tangible potential to reduce water waste, minimize crop losses, and improve the economic sustainability of anthurium cultivation at both smallholder and commercial scales.

10.1 Limitations and Future Work

The current system relies exclusively on API-sourced weather data and does not incorporate on-farm sensor inputs such as soil moisture or canopy temperature, which may further refine irrigation recommendations. Future work should explore the integration of IoT soil sensors to complement API weather data, the development of a feedback mechanism that allows farmers to report crop responses and improve model retraining, the extension of the advisory engine to cover fertilization and pest management decisions, and the adaptation of the core system for additional tropical ornamental crops including Heliconia and Bird of Paradise.

11. BUDGET AND BUDGET JUSTIFICATION

Table 3 --- Budget Table

Description	Cost
Cost for Deployment	10000.00
Internet	4000.00
Travelling cost for data collection	4000.00
Paper Publishing	40000.00
Total	58000.00

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